

COURSE NAME:

STUDENT NAME :

TOTAL HRS :

REG NO :

JOINING DATE :

CONTACT NO. :

STAFF NAME :

Session	Topic		Student Sign	Guide Sign
	Theory	Software		
Session1	Front End HTML: Introductions: What is HTML, HTML Tags HTML Attributes, HTML Images HTML Tables, HTML List, HTML Links, HTML Forms, HTML Block Element, Inline Element, HTML Marquee, HTML iframe, HTML Layout Element, HTML Quotation, HTML Formatting Elements	visual studio code		
Session2	CSS: Introduction: What is CSS? CSS Types: Inline CSS Internal CSS External CSS Color Property Text Property CSS Box Model Background and Border Property Display Property Font Property Padding Margin CSS Navigation Bar Position Property Cursor Property Z-index Property Opacity Property 2D-3D Transform Pseudo-Classes and Pseudo-Elements, Gradient Property Key frame Property	visual studio code		

Session3	BOOTSTRAP: Introduction to Bootstrap framework What is Bootstrap? What is Responsive Web Design? Why Use Bootstrap? Where to Get Bootstrap? Install bootstrap and environment setup Bootstrap CDN CDN Link Bootstrap Tags: Bootstrap Container Bootstrap Grid System Bootstrap Color Bootstrap Table Bootstrap Images Bootstrap Button Bootstrap Progress Bar Bootstrap Spinner Bootstrap Pagination Bootstrap Dropdown Bootstrap button Bootstrap Collapse Bootstrap Navbar Bootstrap Carousel Slideshow Bootstrap Forms	visual studio code		
Session4	JAVA SCRIPT: Introduction JavaScript JavaScript Syntax JavaScript Operators Conditional Statement Looping Statement JavaScript Events JavaScript Object JavaScript Array JavaScript String JavaScript Date Object JavaScript DOM JavaScript Page Redirection JavaScript Popup Boxes Form Validation JavaScript Animation Local storage	visual studio code		

Session5	JQuery Why use jQuery jQuery installation and environment setup jQuery attributes: #ID selector .Classselector jQuery effect: jQuery hide() show () jQuery event: click() mouseenter() mouseleave() mouseup() mousedown() hover() fadeIn() fadeout() fadeToggle() slideUp() slidedown() jQuery animation	visual studio code		
Session1	React ES6 Primer JavaScript history ES6 and Babel Arrow functions and template literals Rest and spread operators Classes and inheritance	Visual Studio Code		
Session 2	Introduction to React Why React? Version history of React React 15 vs React 16 Using create-react-app Debugging first react app	visual studio code		
Session 3	Templating Using JSX Working with React.createElement Logical operators and expressions Specifying attributes and children Function Component Class Component	visual studio code		

Session 4	Components of React Component architecture and its significance Types of components "This" Keyword	visual studio code		
Session 5	Working with Props and State State and its significance Set and read states Passing data to a component through props Validation of props with prop Types Using default props to supply default values to props Rendering lists Using the React key prop Using the map function for iteration on arrays to generate elements	visual studio code		
Session 6	Event Handling in React React event system and the synthetic event Passing arguments to event handlers Component life-cycle	visual studio code		
Session 7	Working with Forms Controlled and uncontrolled components Significance of the default Value prop using the React ref prop to access the DOM element	visual studio code		
Session 8	Routing with React Router Setting up the router Understanding routing with the help of single-page applications Working with HashRouter and BrowserRouter components Configuration of the router, using the route component Using switch component for defining routing rules Working with nested routes Navigation to pages, using NavLink and Link components	visual studio code		

Session 9	Redux and Immutable.js What is Redux? Why Redux? Principles of Redux Installation and setup process of Redux Creation of reducer, actions, and store what is Immutable.js? Collections in Immutable Lists, sets, and maps	visual studio code		
Session 10	React Redux What is React Redux, and why is it used? Installation and setting up of React Redux Container components vs presentational components Understanding the high-order component	visual studio code		
Session 11	React CSS / SASS What is React CSS and SASS, why is it used? Installation and setting up of React SASS	visual studio code		
Session 12	Fragments & Hooks What is Fragment, why is it used? Installation and setting up of React Hooks Programming with Hooks State & Effects	visual studio code		
Session 13	Bootstrap What is Bootstrap, why is it used? Installation and setting up of Bootstrap Css & Styles	visual studio code		
Session 14	Simple Workouts Login Form Using React Sign Up & Sign in Form Using React Create Mail Form Work with Mail Simple App	visual studio code		

	Data Base			
Session1	SQL Server : Introduction of SQL server, SQL server management studio, Create a database, Alter Statement Use data base, Create query, Insert query, Update query, Delete query, Views, Stored Procedures	SQL Server 2022		
Session2	Constraints: Not null constraints, Unique key constraints, Primary key constraints, Foreign key constraints, Default constraints	SQL Server 2022		
	Back End			
Session 1	Introduction To Python : Why to Learn Python? Characteristics of Python Where are Python used? Python Program Structure Applications of Python	(Python) IDLE Shell 3.12		
Session 2	Install Python and Environment Setup: Python 3.12 IDLE	(Python) IDLE Shell 3.12		
Session 3	Basic Concepts: Python Identifiers Python Identifiers Rules Python Keywords Python Variables Creating Variables Input and Output in Python Taking Input from the user Displaying Output Type Casting Get the Type Formatting Output	(Python) IDLE Shell 3.12		

Session 4	Python Basic Data Types: Numbers Sequence Type Boolean , Set ,Dictionary	(Python) IDLE Shell 3.12		
Session 5	Conditional Statement: if statement if-else statement elif statement nested if-else statement pass statement	(Python) IDLE Shell 3.12		
Session 6	Looping Statement: while loop: 1. The while Loop 2. The break Statement 3. The continue Statement 4. The else Statement for loop: 1. Looping Through a String 2. The break Statement 3. The continue Statement 4. The range() Function 5. Else in For Loop 6. The pass Statement	(Python) IDLE Shell 3.12		
Session 7	List: Define list List items: Ordered and Changeable Allow duplicate element List length List item data types Type() ,List() Access list items Negative index Positive indexing Range of index Change list item Change a range of item values Inset items: Append items Extend list Remove list item: Remove specified index Pop() del keyword Clear the list: Clear()	(Python) IDLE Shell 3.12		

Session 8	Tuples: Access tuples: positive indexing negative indexing range of index range of negative index update tuples change tuple values add items: upend() Remove item: remove() del keyword() Loop tuples: 1. for loop through the index numbers: 2. range,len(), Using a while loop: join tuples, join two tuples, multiply tuples, Tuple- methods: Count() index()	(Python) IDLE Shell 3.12		
Session 9	Sets: Define set Len() Set items –data types Types() Access item: For loop in keyword Add set items: add() update() Remove set items: remove() discard() pop() clear() del keyword loop sets: For loop Join two sets: Union Duplicate set: Intersection_update()	(Python) IDLE Shell 3.12		

Session 10	Dictionary: Dictionary items Dictionary length Nested dictionary Type() Get(),Keys(),Values(),Items() Update(),Copy() Remove item: Pop item(),Clear() Dictionary key: Del	(Python) IDLE Shell 3.12		
Session 11	Python Functions: Creating a Function Calling a Function Arguments Number of Arguments	(Python) IDLE Shell 3.12		
Session 12	OOP - Object Oriented Programming: Class Object Abstraction Encapsulation Inheritance: 1. Single Inheritance 2. Multilevel Inheritance 3. Multiple Inheritance 4. Hierarchical Inheritance 5. Hybrid Inheritance Polymorphism Constructor: 1. Default Constructor 2. Parameter Constructor	(Python) IDLE Shell 3.12		
Session 13	Python Modules: Introduction Modules Modules Example	(Python) IDLE Shell 3.12		
Session 14	Python Exception: Introduction Exception Exception Keywords	(Python) IDLE Shell 3.12		
Session 15	File Handling: Working of open() function Mode Of Files OS Built-in Module	(Python) IDLE Shell 3.12		

	Backend Framework (Django)			
Session 1	Introduction: Introduction of Django Django Install ,Django MVT Project Creation ,App Creation	visual studio code		
Session 2	Basic: Django Admin Interface Django Template Layer Django View Functions How to pass the data from views.py to html template 1.List data values 2. Dictionary data values How to insert images How to use Bootstrap , jQuery , font awesome icons , external css	visual studio code		
Session 3	Django Request and Response Object Django Login Page Creation Registration Page Creation	visual studio code		
Session 4	Forms: Django Navigation Django Form Design Django Form Validation Django File Uploading Concept	visual studio code		
Session 5	Database Connection: DB Browser for SQLite SQLite Install	visual studio code		
Session 6	Django CRUD Concept: Create Read Update Delete Django with SQLite	visual studio code		

	REST API			
Session 1	Introduction: Introduction of Rest API REST API Principles	Postman		
Session 2	Basic: HTTP Methods Status code Headers Query params Request Body	Postman		
Session 3	API request API response Async vs Sync	Pycharm		
Session 4	FAST API Setting up FastAPI Basic FastAPI app Pydantic Validation Models SQLAlchemy ORM CRUD Operations	Pycharm		
Session 5	Database Connection: PostgreSQL with FASTAPI	Pycharm		
Session 3	Projects 1: E-commerce Product Catalog (Backend)	Pycharm		
Session 4	Projects 2: Book Library Management (Backend)	Pycharm		
Session 5	Projects 2: Ticket Booking website	Pycharm		

	Full Stack			
Session 1	Introduction: How to connect React with Python API and Database	Vscode Pycharm		
Session 2	Basic: Basic Web App creation User Registration User details update Display user details	Vscode Pycharm		
Session 3	Projects 1: E-commerce Product Catalog (Frontend and Backend)	Vscode Pycharm		
Session 4	Assessment Projects 1: Book Library Management (Frontend and Backend)	Vscode Pycharm		
Session 5	Assessment Projects 2: Ticket Booking website (Frontend and Backend)	Vscode Pycharm		

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